

# 10/9/2026: Contrapositive and contradiction

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CSCI 246: Discrete Structures

Textbook reference: Sec 20, Scheinerman

## Today's Agenda

- Mini lecture ( $\approx 25$  mins)
- Group exercises ( $\approx 25$  mins)

# Overview: Three ways to prove implications

Contrapositive: Details

Contradiction: Details

Applications

There are three ways to prove  $P \implies Q$ .

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2. **By contraposition.** We can prove  $\neg Q \implies \neg P$ .
3. **By contradiction.** We can assume  $\neg(P \implies Q)$  and find a contradiction. (Note: this is the same as assuming  $P \wedge \neg Q$  and finding a contradiction.)

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**By contraposition.** The contrapositive is: if  $n$  is even, then  $5n + 3$  is odd. Let  $n$  be even, so  $n = 2k$ . Then  $5n + 3 = 10k + 3 = 2(5k + 1) + 1$ , which is odd because  $5k + 1$  is an integer. ■

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**By contradiction.** Suppose, for contradiction, that  $5n + 3$  is even but  $n$  is even. Since  $n$  is even, we write  $n = 2k$ . But then  $5n + 3 = 10k + 3 = 2(5k + 1) + 1$ , which is odd. So  $5n + 3$  is both even and odd, which is impossible. ■

Overview: Three ways to prove  
implications

**Contrapositive: Details**

Contradiction: Details

Applications

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**Solution.** Recall that the truth table for the  $\implies$  operator is given by

| P | Q | $P \implies Q$ |
|---|---|----------------|
|   |   |                |
| T | T | T              |
| T | F | F              |
| F | T | T              |
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| P | Q | $\overbrace{P \implies Q}^{\text{If P, then Q}}$ |
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| T | T | T                                                |
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Now what?

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| F | F | T                                                |

Now we apply the  $\implies$  operator to inputs transformed by the "not" operator (also written  $\neg$ ).

| P | Q | $\neg Q$ | $\neg P$ | $\overbrace{\neg Q \implies \neg P}^{\text{If not Q, then not P}}$ |
|---|---|----------|----------|--------------------------------------------------------------------|
| T | T | F        | F        | T                                                                  |
| T | F | T        | F        | F                                                                  |
| F | T | F        | T        | T                                                                  |
| F | F | T        | T        | T                                                                  |

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| P | Q | $\overbrace{P \Rightarrow Q}^{\text{If P, then Q}}$ |
|---|---|-----------------------------------------------------|
| T | T | T                                                   |
| T | F | F                                                   |
| F | T | T                                                   |
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|---|---|----------|----------|-----------------------------------------------------------------------|
| T | T | F        | F        | T                                                                     |
| T | F | T        | F        | F                                                                     |
| F | T | F        | T        | T                                                                     |
| F | F | T        | T        | T                                                                     |

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|---|---|----------|----------|--------------------------------------------------------------------|
| T | T | F        | F        | T                                                                  |
| T | F | T        | F        | F                                                                  |
| F | T | F        | T        | T                                                                  |
| F | F | T        | T        | T                                                                  |

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Hence  $P \implies Q$  is equivalent to  $\neg Q \implies \neg P$ .

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**Claim.** Let  $a, b$  be integers. If  $ab$  is even, then  $a$  is even or  $b$  is even.

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**Proof.** Write  $a = 2k + 1$  and  $b = 2\ell + 1$ . Then

$$\begin{aligned} ab &= (2k + 1)(2\ell + 1) = 4k\ell + 2k + 2\ell + 1 \\ &= 2(2k\ell + k + \ell) + 1, \end{aligned}$$

which has the form  $2m+1$ , i.e. odd. So the original claim holds.

Overview: Three ways to prove  
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Contrapositive: Details

**Contradiction: Details**

Applications

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| P | Q | $\overbrace{P \implies Q}^{\text{If P, then Q}}$ | $\neg(P \implies Q)$ |
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**Conclusion.** We can prove  $P \implies Q$  by assuming  $P \wedge \neg Q$  and finding a contradiction.

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Now:

- If  $n$  is odd, write  $n = 2k + 1$ .
- Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd.  $\Rightarrow \Leftarrow$

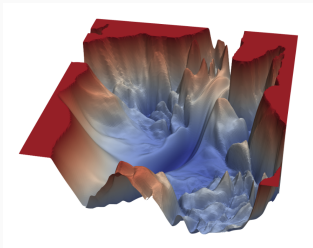
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**Applications**

# Machine learning: Why gradient descent stops

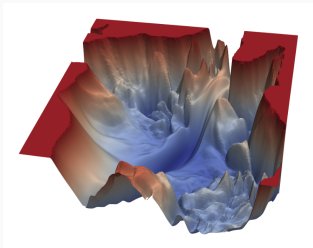


A nonconvex loss surface for a neural network: many local minima and saddle points where gradient descent can stop.

Source: H. Li, Z. Xu, G. Taylor, C. Studer, & T. Goldstein, "Visualizing the Loss Landscape of Neural Nets," *NeurIPS* 2018.

**Fact.** If  $x$  is a local minimum of a differentiable function  $f$ , then  $\nabla f(x) = 0$ .

# Machine learning: Why gradient descent stops



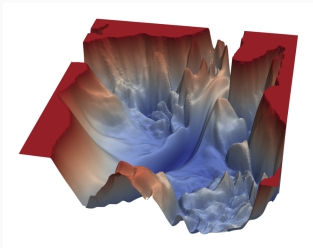
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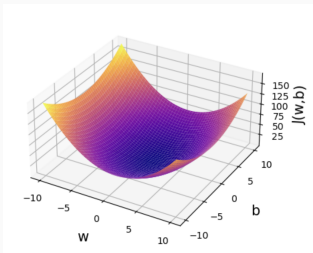
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**Remark.** The second form is the version gradient descent actually uses: keep stepping while the gradient is nonzero.

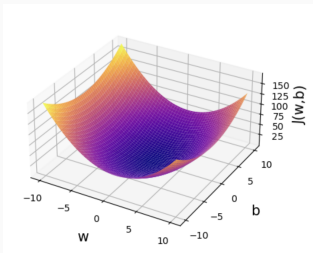
# Machine learning: Convex losses have no bad local minima



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**Claim.** If a function  $f$  is convex, every local minimum of  $f$  is a global minimum.

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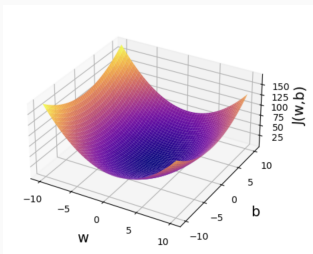


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**Claim.** If a function  $f$  is convex, every local minimum of  $f$  is a global minimum.

**Proof sketch.** By contradiction. Suppose  $x^*$  is a local minimum but not a global one. Then some  $y$  has  $f(y) < f(x^*)$ . Convexity says  $f$  lies below the chord on the segment from  $x$  to  $y$ , so points on that segment arbitrarily close to  $x^*$  have values below  $f(x)$ . That contradicts  $x$  being a local minimum. ■

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**Who cares.** Linear regression and logistic regression have convex losses. This short contradiction argument guarantees that gradient descent can't get stuck at a bad solution on those models.

## **Group exercises**